



CS355 - Digital Forensics

Report for Labs 6-8

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1. Lab 6

1.1. Exercise 1

- (d) Each image provided demonstrates a spliced image involved. Firstly with the 'tiger.jpg' file, we can see from the image at quality compression 78[1], which shows the outline of the tiger from the original image. We can clearly see it outlined very obviously, and when comparing it to the original image, we can deduce the tiger was spliced in.

Next looking at 'soldier.jpg', we can identify that there are some spliced stripes present on the soldier's arm, by looking at the difference image using a quality of 93[2]. We can then compare to the original image to see the stripes are likely the flags and badges of the soldier which has been spliced on.

Finally 'beach.jpg', we can identify that the girl has an outline of white spots around her, which suggests they were edited and spliced in, mostly clearly seen at quality 65[3].

2. Lab 7

2.1. Exercise 1

- (a) Here is the binary mask generated by blockMatch with $t = 0.005$ on the original image[4].
- (b) Here is the binary mask generated with the fewest false positives possible[5]. $t = 0.013$ was the decided threshold; with the only false positives being three small spots in the trees above, and the plane empty ground below. Looking at the attention mask we can see that two share an identical shape, implying they are copies, while the three spots, and the larger spot are all unique. They are likely a result of similar blocks nearby due to the compression now being unable to distinguish them.

3. Lab 8

Project presentation really unfortunately aligned with this due date; hence lab 8 was not completed.

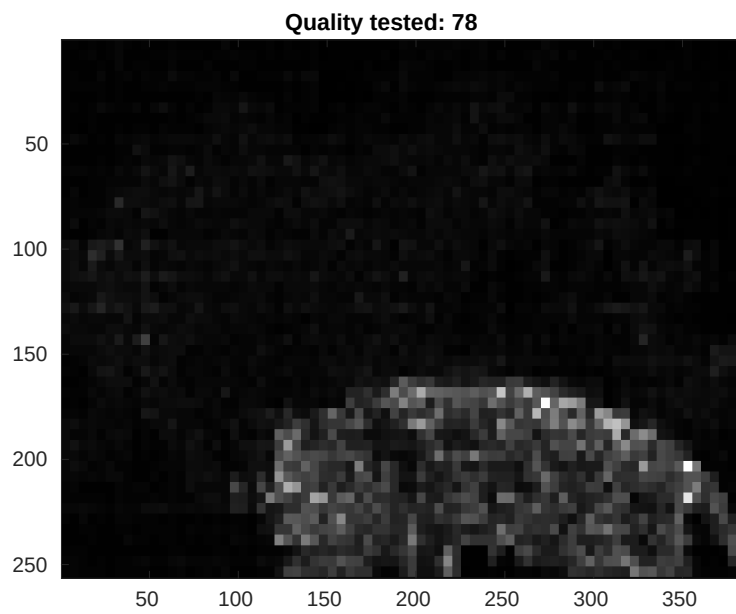


Fig 1: JPEG Ghosts in 'tiger.jpg'.

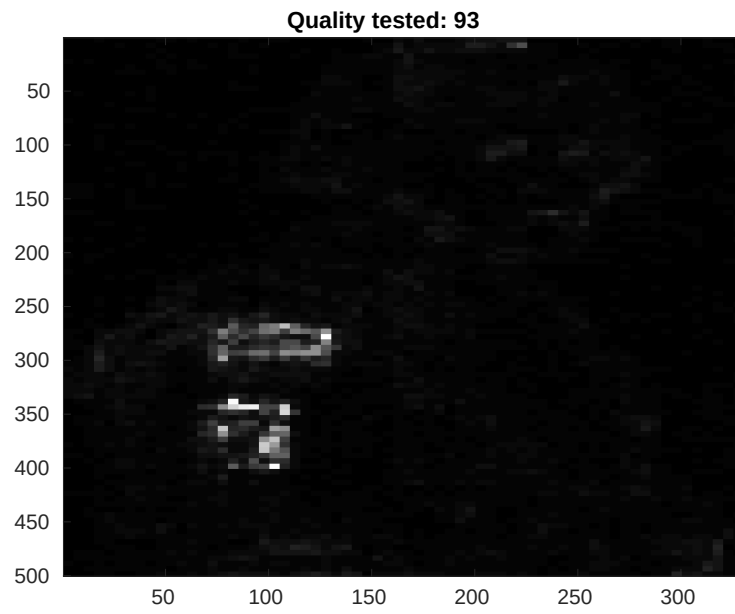


Fig 2: JPEG Ghosts in 'soldier.jpg'.

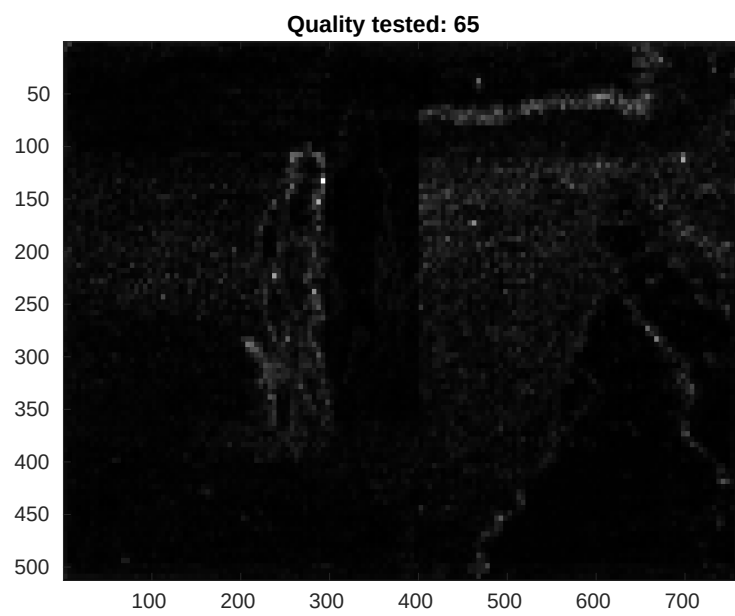


Fig 3: JPEG Ghosts in 'beach.jpg'.

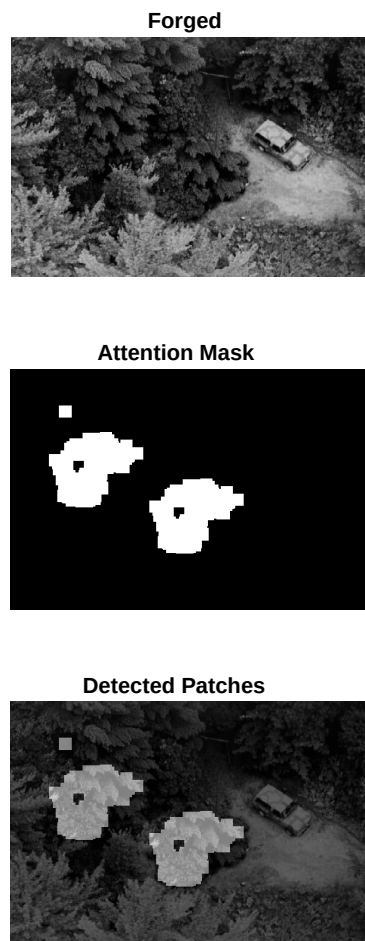


Fig 4: Binary mask depicting copy and pasted regions, threshold $t = 0.005$.

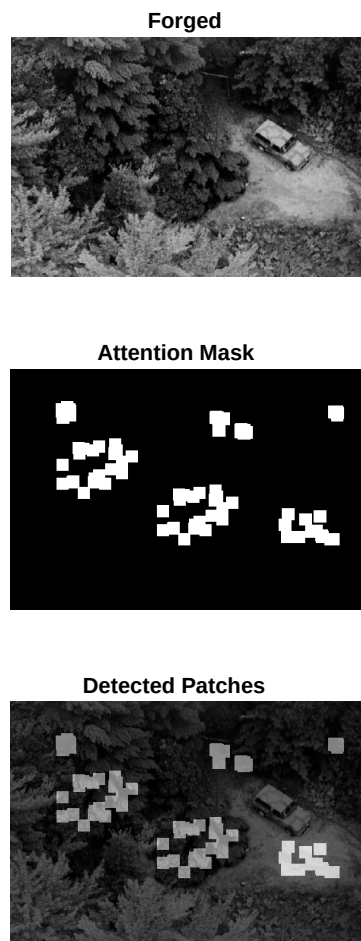


Fig 5: Binary mask depicting copy and pasted regions, threshold $t = 0.013$ on quality 90.